Condition Index: How to Normalize

Turn any indicator into a 0 – 1 score

HIGHER IS BETTER ▲

$$CI = Value / Reference$$

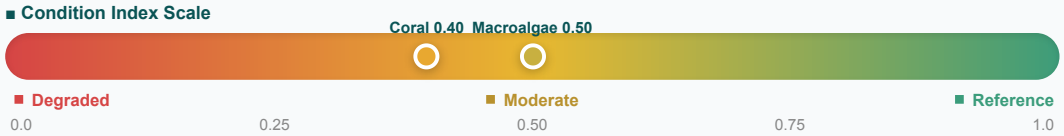
Use when **more** of the indicator means better condition.
Examples: coral cover, fish biomass, species richness.

HIGHER IS WORSE ▼

$$CI = 1 - (Value / Reference)$$

Use when **more** of the indicator means worse condition.
Examples: macroalgae cover, turbidity, nutrient loading.

■ Condition Index Scale



Higher-is-better

◈ Coral Cover

Observed: **20%** | Reference: **50%**

$$CI = 20 / 50 = 0.40$$

Below moderate — degraded condition

Higher-is-worse

▲ Macroalgae Cover

Observed: **30%** | Reference threshold: **60%**

$$CI = 1 - (30 / 60) = 0.50$$

Moderate condition

- ✓ **Cap CI** between 0 and 1
- ✓ **State direction** for every indicator
- ✓ **Consistent references** across time
- ✓ **Document** reference source